An AR-based Approach for Visualizing Indoor Information

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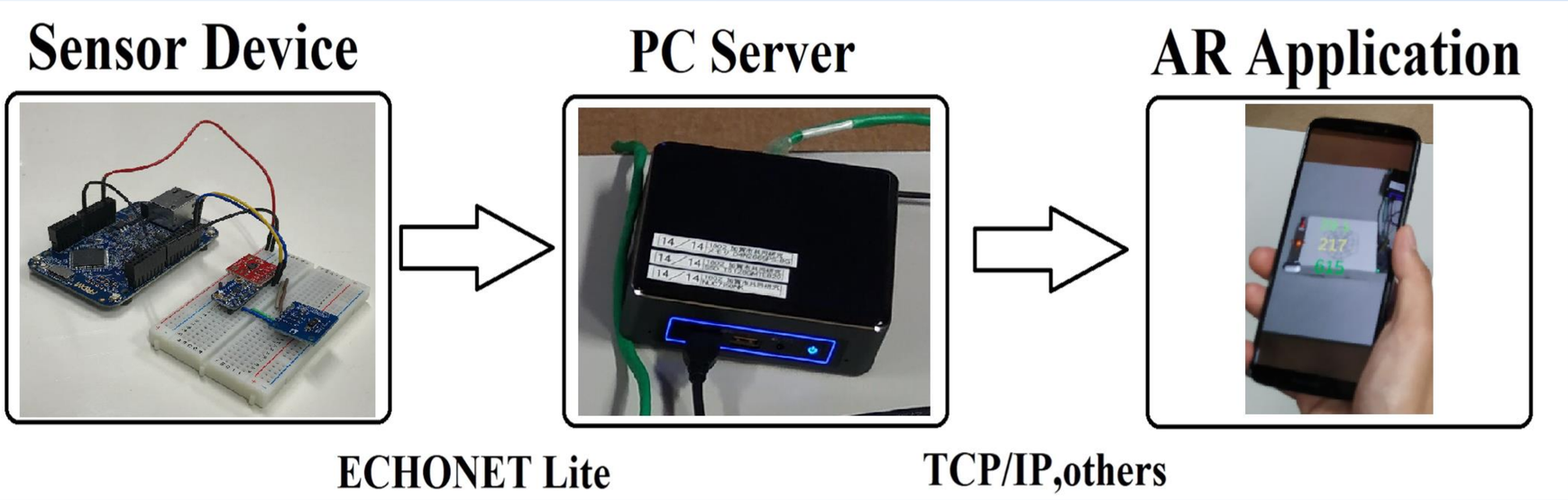
1. Introduction

Comfortable and safe indoor spaces are necessary for a good quality of living. To achieve such indoor spaces, the first step is to gather sensor data such as temperature, illumination and carbon dioxide. Such solutions usually produce graphs and visualizations that are not easily understood by non-expert users. To this end, we developed a visualization system based on AR technology so that data can be used intuitively for everyone.



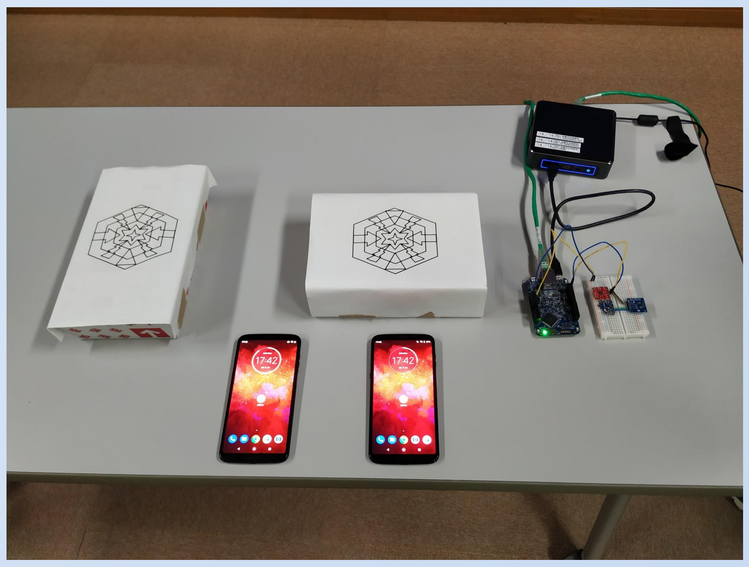


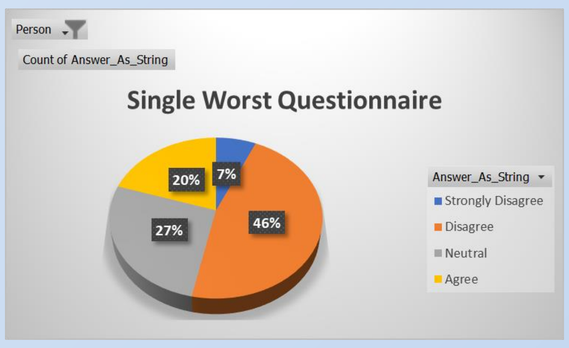
Fig. 1: System Design (left), Experimental Setup (right)

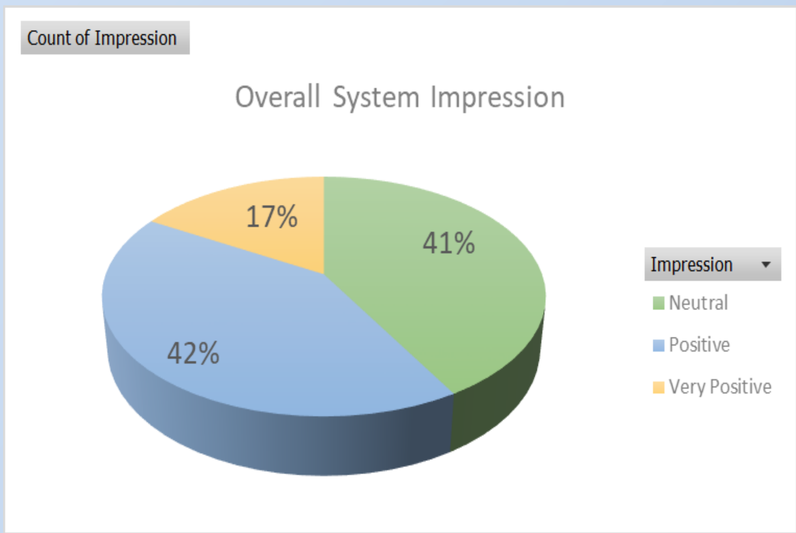
2. System Design

The system uses a network communication protocol called ECHONET Lite[1] to transfer data from sensors to a server and display this information on a mobile phone. The phone app uses a game engine (software that performs real-time processing of images and sounds) called Unity[2] (Fig.1). Unity uses the AR Foundation framework to display data in space by reading markers.

3. Experiments

To evaluate the system, an experiment session was organized (Fig. 2). 12 people interacted with the system and filled a questionnaire. After analyzing the results, 59% of the respondents had a positive impression and 41% were neutral. Average question scores, as well as the best and worst user evaluations are presented in the following figures.





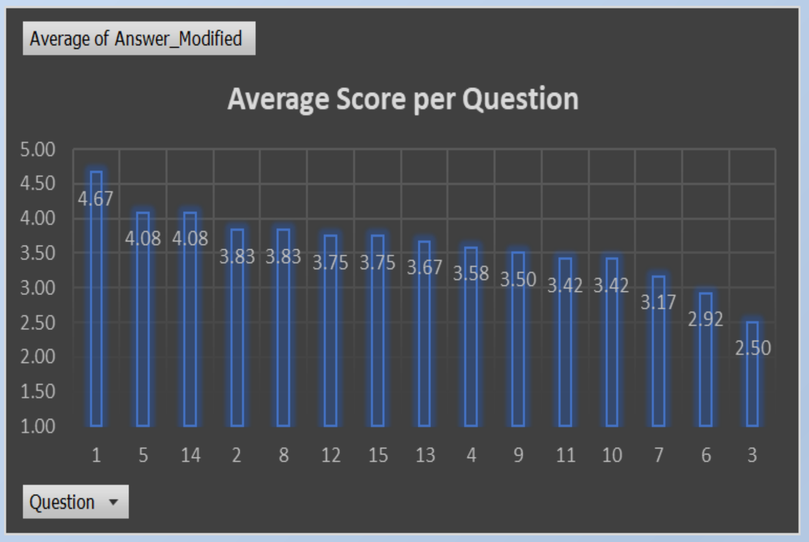


Fig. 2: Experiment (left), Best and Worst evaluations (middle),
Overall System Impressions (top right) Average Question Scores (bottom left)

4. Results and Observations

The respondents indicate that the information is simple and easy to understand. Furthermore, the responsiveness of the system was evaluated favourably. In contrast, the 3D graphics used in the application received a neutral response at best and many users reported that the lack of measuring units make the information difficult to understand. A few respondents reported that their eyes get tired easily. Most importantly, many users reported that they do not know what to do next, using that information.

5. Future Prospects

We were able to provide information simply, but we still need to adjust the visualization. Furthermore, since people reported that they did not know what to do with the information, a suggestion system that presents the user with the next step should be introduced. For example, "Let's open the window." when the temperature or carbon dioxide concentration rises, or "Let's turn on the light." when the illuminance decreases.

6. References [1] ECHONET: https://echonet.jp/ [2] Unity: https://unity.com/